Notes: Acharya & Rajan (JF, 2024)

Liquidity, Liquidity Everywhere, Not a Drop to Use: Why Flooding Banks with Central Bank Reserves May Not Expand Liquidity

Contents

1 Big picture

- **Question.** Why can large central-bank reserve balances coexist with recurring liquidity stress in money markets, banks, and credit markets?
- **Core claim.** Central-bank reserves are created through commercial banks. When banks finance those reserves with demandable deposits and other liquidity promises, reserve creation also creates future claims on liquidity.
- **Main mechanism.** Quantitative easing raises bank reserves, but it can also raise uninsured demandable deposits and credit-line commitments. If reserves later shrink, become encumbered, or are hoarded by healthy banks, stressed banks may face a larger liquidity shortage.
- **Key friction.** Healthy banks with spare reserves may prefer to remain visibly safe and attract flight-to-safety deposits rather than lend reserves in the interbank market. This can keep reserves from flowing to banks that need them.
- **Counterintuitive result.** Under high liquidity stress, reserve shrinkage, and low interbank participation, increasing reserves at date 0 can increase the date-1 interbank rate premium.
- **Real effect.** A higher anticipated interbank premium raises banks' date-0 loan rates, reduces firm investment, and weakens the real-activity benefits of balance-sheet expansion.
- **Policy implication.** A central bank balance sheet expansion is not equivalent to net liquidity creation unless the associated private liquidity claims, reserve encumbrance, and hoarding incentives are also controlled.
- **Main warning.** Repeated QE and rescue interventions can ratchet up private liquidity dependence, forcing larger future interventions and creating a form of financial dominance.

2 Data, variables, and institutional motivation

2.1 Why the empirical motivation matters

The paper is primarily theoretical, but it is motivated by repeated episodes in which supposedly ample reserves did not prevent liquidity stress: the U.S. repo-market spike in 2018–2019, March 2020 market stress, and the March 2023 regional-bank episode.

Interpretation: The empirical puzzle is not simply that there were too few reserves in an accounting sense. The puzzle is that reserves were large relative to pre-crisis levels, yet market-clearing liquidity was still scarce when many claims on liquidity were called simultaneously.

2.2 Reserve expansion through bank balance sheets

- Nonbanks generally cannot hold central-bank reserves directly.
- If the central bank buys assets from nonbanks, nonbanks deposit the payment into commercial banks.
- The banking system therefore receives both a reserve asset and a deposit liability.
- In aggregate, banks cannot make the reserves disappear; they can only change the liability mix used to finance them.

Interpretation: The balance-sheet channel is central. A dollar of reserves held by banks is not a free dollar of systemic liquidity if it is matched by a demandable claim that may be withdrawn during stress.

2.3 Key state variables

- S_0: aggregate reserves held by banks at date 0.
- τ: fraction of date-0 reserves that shrink or become encumbered by date 1.
- θ: fraction of bank-firm pairs stressed when the economy is in the liquidity-stressed state.
- φ: fraction of healthy banks that lend in the interbank market and become "tainted."
- δ: convenience yield from holding reserves in the stressed state.
- r_1: interbank-rate premium or, when the interbank market is shut, the stressed bank's autarky marginal funding premium.

Interpretation: The model is about *net usable liquidity*: reserves available to stressed banks after subtracting reserve shrinkage, liquidity claims, and hoarded reserves.

2.4 Why deposits matter

- Banks often finance reserve balances with short-term demandable deposits, including uninsured deposits.
- Deposits are liquidity claims. In stress, depositors may withdraw from troubled banks and move to safe banks.
- Deposits that financed reserves at date 0 need not shrink one-for-one when reserves shrink later.
- This creates a mismatch: the liquidity claims remain, but the reserve assets backing them may be lower or inaccessible.

Interpretation: The same reserve expansion that looks stabilizing from the asset side can create fragility from the liability side.

2.5 What Exhibit 1 contributes

Exhibit 1 is the paper's empirical motivation. It shows the time-series behavior of reserves, deposits, credit lines, and the ratio of liquidity claims to reserves.

Interpretation: The exhibit is not a causal test. It is a disciplined fact pattern motivating the model: reserves grew, but claims on liquidity also grew, so the relevant ratio is not reserves alone but reserves relative to demandable claims and commitments.

3 Model structure and main equations

3.1 Timeline and states

The economy has dates 0, 1, and 2. At date 0, firms and banks make financing and investment decisions. At date 1, the economy may be healthy or liquidity stressed. If stressed, some firms and their banks need liquidity for rescue investment. Healthy banks may either lend in the interbank market and become tainted or remain safe and attract flight-to-safety deposits.

Interpretation: The model is deliberately simple: it focuses on the liquidity consequences of reserve expansion, not on the direct effects of QE on asset prices, monetary stimulus, or aggregate demand.

3.2 Firm problem

At date 0 and date 1, the firm solves

$$\max_{L_0^F, D_0^F} \quad (1-q)[g_0(I_0) + D_0^F] + q[g_1(I_1) - l_1^F(1+\gamma+r_1)] - R_0^L L_0, \tag{1}$$

$$\max_{l_1^F} \quad g_1(I_1) - l_1^F(1+\gamma+r_1), \tag{2}$$

subject to

$$I_0 = L_0 + W_0^F - D_0^F, \qquad I_1 = l_1^F + D_0^F. \tag{3}$$

The first-order conditions are

$$(1-q)g_0'(I_0) - R_0^L = 0, \tag{1}$$

$$(1-q)(-g_0'(I_0) + 1) + qg_1'(I_1) = 0, \tag{2}$$

$$g_1'(I_1) - (1+\gamma+r_1) = 0. \tag{3}$$

Interpretation: The firm holds precautionary deposits because deposits reduce the need to borrow in the stressed state. If the date-1 liquidity premium r_1 rises, rescue investment I_1 becomes more expensive and date-0 lending terms change.

3.3 Bank problem and capital issuance

The bank funds date-0 loans and reserves with deposits and capital. It faces convex costs of issuing capital at dates 0 and 1. The key date-1 first-order condition is

$$r_1 = \alpha_1 e_1. \tag{8}$$

The rate relation is

$$R_0^L = R_0^{DF} = 1 + q\gamma + qr_1 = (1+\lambda L_0)(1+qr_1). \tag{9}$$

Interpretation: The interbank premium transmits directly into bank capital issuance, bank loan pricing, and firm investment. Liquidity stress is therefore not only a date-1 financial-market issue; it affects date-0 real activity through expectations.

3.4 Market clearing and net liquidity shortfall

Define the net shortfall function

$$f(r_1, S_0) = \frac{\theta}{\varphi(1-\theta)+\theta}\left[I_1 + (D_0 - D_0^F)\right] - S_0(1-\tau). \tag{10}$$

If $f(0, S_0) \le 0$, the interbank premium is zero. If $f(0, S_0) > 0$, the premium is positive and capital issuance must rise to clear the stressed liquidity market.

Interpretation: The first term is the liquidity demand per participating stressed/surplus bank pool. The second term is usable reserves after shrinkage. The expression highlights that reserves are useful only to the extent they remain available and reach stressed banks.

3.5 Theorem 1: when more reserves raise the interbank premium

The model's central result is

$$\frac{d\bar{r}_1}{dS_0} > 0 \quad \Longleftrightarrow \quad \theta > \frac{\varphi(1-\tau)}{\tau + \varphi(1-\tau)}. \tag{T1}$$

Interpretation: The condition says that reserves can be counterproductive when marginal liquidity demand from stressed banks exceeds marginal usable liquidity supplied to stressed banks. This is more likely when stress is widespread (θ high), reserves shrink or are encumbered (τ high), and few healthy banks lend in the interbank market (φ low).

Policy interpretation: In the traditional view, reserves are always a buffer. In this model, reserves are a buffer only if they are not offset by runnable claims and are actually redistributed to the banks that need them.

3.6 Endogenous interbank participation and reserve hoarding

When healthy banks choose whether to lend in the interbank market, φ is endogenous. The indifference condition between lending and staying safe can be written as

$$1 - \varphi = \frac{\delta S_0(1-\tau)}{(1-\theta)\left[r_1 S_0(1-\tau) + r_1^2/(2\alpha_1)\right]}. \tag{13}$$

Interpretation: The left side is the share of healthy banks that stay safe. The right side rises when the convenience yield on safety is high. A higher stock of reserves can make remaining safe more attractive because safe banks may attract more flight-to-safety deposits.

3.7 Theorem 2: three regimes with endogenous hoarding

With $\delta > 0$ and $\tau > 0$, there are two thresholds S_0^* and S_0^{**}.

- For $S_0 \leq S_0^*$, stressed banks are not liquidity deficient and the rate charged to firms is δ.
- For $S_0 \in (S_0^*, S_0^{**}]$, stressed banks are liquidity deficient, but the interbank market remains shut; stressed banks are in autarky.
- For $S_0 > S_0^{**}$, the interbank market opens; the equilibrium rate satisfies $\bar{r}_1(S_0) \geq r_1^{\varphi}(S_0) > 0$.

Interpretation: More reserves need not immediately open the interbank market. There can be an intermediate range where stressed banks become more liquidity deficient but healthy banks still prefer to hoard.

3.8 Theorem 3: convenience yield worsens hoarding

The paper shows that S_0^* and S_0^{**} increase in δ, and when the interbank market is open,

$$\frac{d\bar{r}_1}{d\delta} > 0. \tag{T3}$$

Interpretation: A higher convenience yield makes safe status more valuable. This shifts the threshold for interbank participation to the right and raises the stressed-state premium. The market can be full of reserves, but if reserves are prized as safety badges, they are not necessarily lent.

3.9 Reserves shrinkage and QT

The model interprets τ broadly. It can arise from quantitative tightening, reserve encumbrance, or reserves being pledged or otherwise unavailable in stress.

- QT can reduce bank-held reserves without reducing the deposits created during QE.
- Encumbrance can reduce usable reserves even if measured reserve balances remain high.
- If a higher share of reserves is unavailable at date 1, the conditions for stress become easier to satisfy.

Interpretation: The problem is not only the level of reserves but the mismatch between reserve balances and liquidity claims after the central-bank balance sheet evolves.

3.10 Bank capital structure and ex post intervention

The paper also studies whether capital requirements and emergency central-bank lending solve the problem. With endogenous hoarding, the planner may not simply want banks to hold more date-0 capital, because more capital can lower the interbank rate and strengthen incentives for safe banks to hoard. Ex post intervention can reduce the premium, but it may crowd out private interbank lending and create expectations of future support.

Interpretation: The model is a warning against relying on one instrument. More reserves, more capital, and more emergency intervention can each change private incentives in ways that offset the intended liquidity effect.

4 Identification and conceptual design

4.1 What kind of paper this is

This is a theoretical paper with empirical motivation. It does not estimate a treatment effect. Instead, it provides a mechanism that rationalizes why reserve abundance need not eliminate liquidity stress.

Interpretation: The model's credibility comes from internal logic and from its ability to organize observed episodes: QE, QT, deposit creation, flight-to-safety flows, interbank-market fragility, and emergency central-bank interventions.

4.2 What the model isolates

- It holds aside the direct stimulus effects of asset purchases.
- It focuses on the commercial-bank balance-sheet channel.
- It emphasizes endogenous private liquidity claims and endogenous hoarding.
- It studies how expected future liquidity stress feeds back into current lending and investment.

Interpretation: The central analytic move is to compare gross reserves with net reserves after subtracting the private claims and behavior induced by reserve creation.

4.3 Why the mechanism is plausible

- Banks issue demandable liabilities because deposits are valuable and cheap.
- Demandable liabilities are claims on liquidity when stress arrives.
- Healthy banks may care about being perceived as safe.

- Interbank lending can taint the lender by revealing reserve use or exposing it to repayment risk.
- Central-bank support may induce further private liquidity promises.

Interpretation: These mechanisms are all individually familiar. The paper's contribution is showing their interaction: reserve expansion can create new liquidity claims, and the distribution of reserves can become distorted exactly when liquidity is needed.

4.4 What the model does not claim

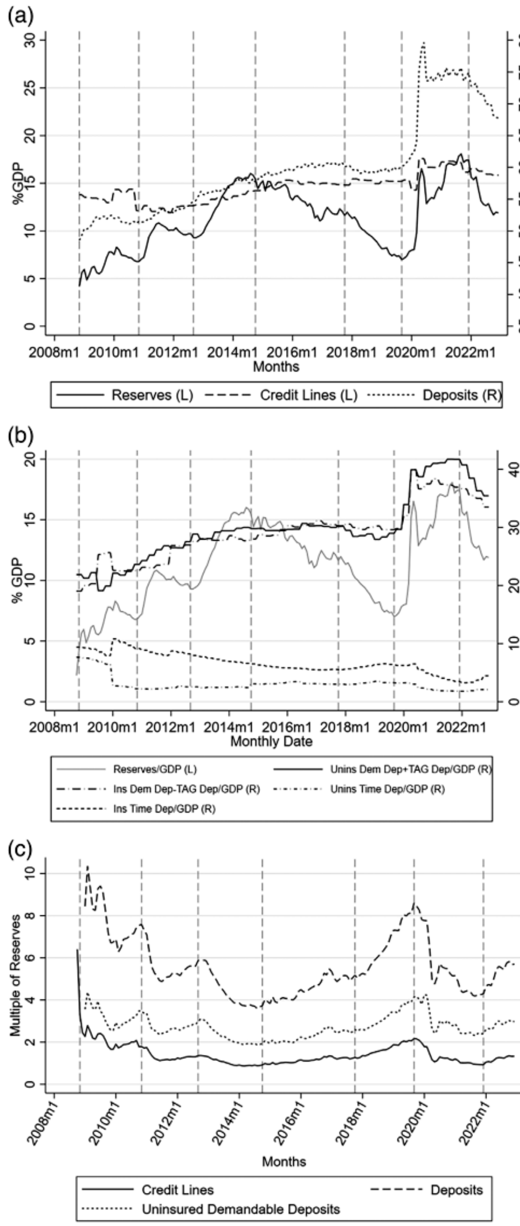
- It does not say QE always worsens liquidity.
- It does not deny that emergency central-bank liquidity can be necessary.
- It does not claim reserves are irrelevant.
- It does not model every regulatory constraint or every money-market institution.
- It does not provide an empirical estimate of the optimal central-bank balance sheet.

Interpretation: The paper's claim is conditional: reserves can fail to expand net liquidity when the banking system responds by issuing liquidity claims and hoarding reserves in stress.

5 Tables and figures

The paper has no empirical tables. It has one empirical motivation exhibit and several model diagrams. The visuals below are directly cropped from the paper and grouped compactly where possible. I add more detailed interpretation under each visual to clarify what each panel contributes to the model.

5.1 Exhibit 1: time-series of credit lines, deposits, and reserves



Original Exhibit 1: time-series of credit lines, deposits, and reserves.

Read:

- Panel (a) plots reserves, credit lines, and deposits as a percentage of GDP.
- Reserves rise sharply around major QE episodes and especially around the pandemic period.
- Deposits also rise substantially and remain large relative to GDP.
- Panel (b) decomposes deposit claims into insured/uninsured and demand/time components, showing that

uninsured demandable deposits are an important part of the liquidity-claim stock.

- Panel (c) normalizes liquidity claims by reserves. The ratio of deposits and uninsured demandable deposits to reserves rises especially when reserves decline after prior QE phases.

Interpretation: The exhibit changes the unit of analysis from "how many reserves exist" to "how many liquidity claims exist per dollar of reserves." This is exactly the paper's central move. Even if reserves are high in levels, the banking system may be fragile if deposits, credit lines, and uninsured demandable claims grow faster or fail to shrink when reserves shrink.

Economic message: The key empirical interpretation is that QE can leave behind a larger liability structure. During QT, reserves fall first, while deposits and commitments may decline slowly. That creates a higher claims-to-reserves ratio and helps explain why liquidity stress can appear even when reserves look ample relative to pre-2008 norms.

Policy interpretation: Exhibit 1 suggests that central banks should monitor a broader liquidity-balance-sheet dashboard: reserves, uninsured demandable deposits, credit lines, reserve encumbrance, and the distribution of reserves across safe and stressed banks. A reserve quantity alone is too narrow.

5.2 Figure 1 and Figure 2: model states and balance sheets

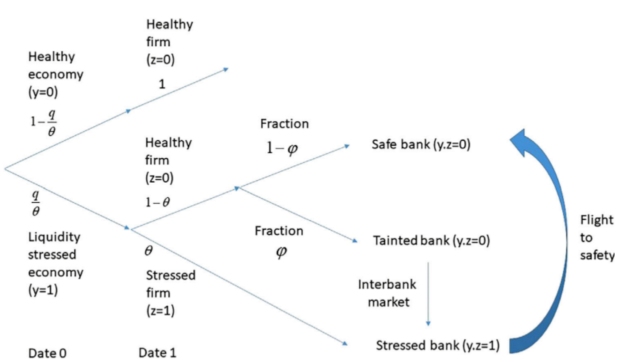


Figure 1. Time line and state space of outcomes. (Color figure can be viewed at wileyon-

(a) Figure 1: timeline and state space.

Firm Balance Sheet at Date 0

Assets	Liabilities
I_0	$L_0^F \quad (= L_0^B)$
D_0^F	W_0^F
	Net worth

Bank Balance Sheet at Date 0

Assets	Liabilities
$L_0^B + \frac{1}{2}\lambda(L_0^B)^2$	D_0
S_0	e_0
	Net worth

Firm Balance Sheet at Date 1 if stressed

Assets	Liabilities
I_1	l_1^F
	L_0^F
	Net worth

Bank Balance Sheet at Date 1 if bank stressed

Assets	Liabilities
$L_0^B + \frac{1}{2}\lambda(L_0^B)^2$	Possible interbank borrowing = b_1
τS_0	e_1
$l_1^B \quad (= l_1^F)$	e_0
	Net worth

Firm Balance Sheet at Date 1 if healthy

Assets	Liabilities
I_0	L_0^F
D_0^F	W_0^F
	Net worth

Bank Balance Sheet at Date 1 if economy stressed, bank healthy but "tainted" (makes interbank loans)

Assets	Liabilities
$L_0^B + \frac{1}{2}\lambda(L_0^B)^2$	D_0
Interbank loans of up to $e_1 + (1-\tau)S_0$	e_1
Reserves of ($S_0 + e_1$ -interbank loans)	e_0
	Net worth

Figure 2. Bank and firm balance sheets.

(b) Figure 2: firm and bank balance sheets.

Read: Figure 1:

- The economy can be healthy or liquidity stressed.
- In the stressed economy, only a fraction θ of firms and banks are stressed.
- Healthy banks choose whether to lend in the interbank market and become tainted or stay safe.
- Safe banks attract flight-to-safety flows from stressed or tainted institutions.

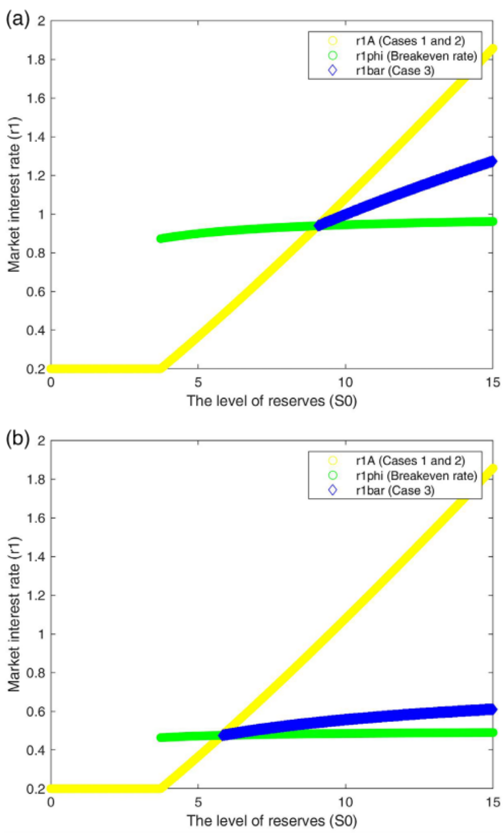
Read: Figure 2:

- At date 0, the bank holds loans and reserves, financed by deposits and capital.
- At date 1, a stressed bank has shrunken reserves τS_0, firm rescue lending, deposit outflows, and possible interbank borrowing.
- A healthy tainted bank lends in the interbank market using reserves and capital.
- A healthy safe bank avoids interbank lending and benefits from flight-to-safety inflows.

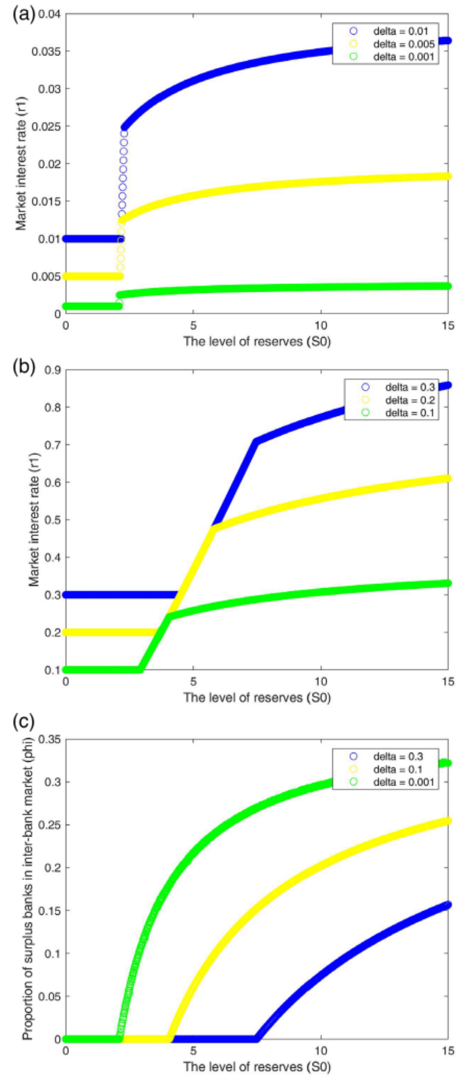
Interpretation: These two figures together show the paper's key strategic environment. Liquidity is not merely a quantity; it is a network allocation problem. Reserves must move from safe banks to stressed banks, but the safe banks may prefer not to move them. The balance-sheet diagram then shows why the stressed bank's problem is real: deposits and rescue lending create cash needs, while only a fraction of initial reserves may be usable.

Economic message: The figures make the model's core logic visible: reserve expansion can load banks with liquid assets and demandable liabilities at the same time. The liability side can become a claim on the asset side during stress, and the asset side may not travel to where it is needed.

5.3 Figure 3 and Figure 4: endogenous interbank-market opening and reserve hoarding



(a) Figure 3: varying stress intensity θ.



(b) Figure 4: varying the convenience yield δ.

Read: Figure 3:

- The yellow line is the autarkic rate that stressed banks would face if the interbank market stayed closed.
- The green line is the breakeven rate required to induce healthy banks to enter the interbank market.
- The blue line is the actual equilibrium rate when the interbank market opens.
- In both panels, the equilibrium rate rises with reserves over a range of S_0.
- The interbank market opens only after the autarky rate exceeds the breakeven rate.

Read: Figure 4:

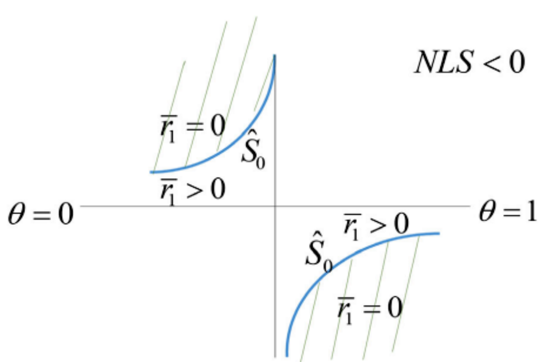
- Panels (a) and (b) show that a higher convenience yield shifts the relevant thresholds and raises the market rate.
- Panel (c) shows that higher δ lowers φ, the fraction of healthy banks willing to lend in the interbank market.
- When safety is valuable, more healthy banks prefer to stay safe and collect flight-to-safety deposits.

Interpretation: Figure 3 is the clearest numerical illustration of the paper's counterintuitive result. More reserves can raise the stressed-state rate because reserves also finance deposits and future liquidity claims. Figure 4 explains why the redistribution channel can break: the more valuable it is to look safe, the fewer healthy banks lend reserves to stressed banks.

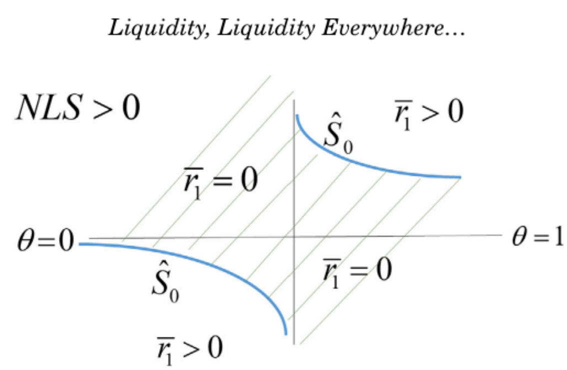
Economic message: The figures distinguish *gross reserve abundance* from *market-clearing liquidity*. The banking system may have many reserves in aggregate, yet the price of liquidity can rise if reserves are held by banks that prefer hoarding. This is why the paper's mechanism is not only about aggregate scarcity but also about allocation and incentives.

Policy interpretation: The policy interpretation is subtle. Simply raising S_0 may move the economy along the x-axis into a region with higher rates if deposits and hoarding adjust endogenously. Reducing δ-type hoarding incentives, reducing reserve encumbrance τ, or improving interbank-market participation φ may be more effective than only increasing gross reserves.

5.4 Figures A1 and A2: threshold logic behind Theorem A1



(a) Figure A1: threshold when $NLS < 0$.



(b) Figure A2: threshold when $NLS > 0$.

Read:

- The vertical axis is reserves and the horizontal axis is the stress share θ.
- The threshold $\widehat{S}_0$ separates regions with zero versus positive interbank premia.
- The figures show that the sign of the reserve effect can switch across parameter regions.
- In the unconventional region, moving to higher reserves can move the economy into a positive-rate region rather than away from it.

Interpretation: The appendix figures provide the geometry behind Theorem 1. They show that the traditional intuition is only one region of the parameter space. Once reserve shrinkage and incomplete interbank-market participation matter, the zero-premium region can shrink or shift in ways that make more reserves insufficient, or even harmful at the margin.

Economic message: The deeper message is that reserve policy has thresholds. A policy that works in one region can fail in another. This helps explain why central banks may believe they hold “ample” reserves, yet market stress reappears after changes in the liability structure, reserve distribution, or QT.

6 Short summary

- The paper studies why large central-bank reserve balances may fail to eliminate liquidity stress.
- The key reason is that reserves are intermediated through commercial banks.
- Banks finance reserves with deposits and other liquidity claims, especially uninsured demandable deposits.
- During stress, those claims can be called, while reserves may shrink, be encumbered, or be hoarded by healthy banks.
- The date-1 interbank premium affects date-0 lending, bank capital issuance, and firm investment.
- Theorem 1 shows that more reserves can raise the interbank premium when stress is widespread, reserve shrinkage is high, and interbank participation is low.
- Endogenous hoarding makes the problem sharper because healthy banks may prefer to remain safe rather than lend.
- A higher convenience yield on reserves shifts the interbank-market opening threshold upward and raises the stressed-state rate.
- Exhibit 1 motivates the model by showing that claims on liquidity can grow relative to reserves.
- The policy lesson is that central banks should track net liquidity, not just the reserve quantity.

7 Conclusion

7.1 Core conclusion

Acharya and Rajan argue that central-bank balance sheet expansion can create less net liquidity than the raw reserve quantity suggests. The reason is that reserves are not injected into a frictionless market. They are issued through commercial banks, whose liability choices and hoarding incentives determine whether reserves are available to meet future liquidity demand.

7.2 Economic mechanism

The mechanism has four linked steps. First, the central bank buys assets and creates reserves. Second, commercial banks end up holding those reserves and often finance them with demandable deposits. Third, reserves can shrink or become unavailable later, while the deposits and credit-line claims remain. Fourth, when stress arrives, safe banks may prefer to hoard reserves and attract flight-to-safety inflows instead of lending to stressed banks. The system can therefore be rich in reserves but poor in transferable liquidity.

7.3 Why this changes how to think about QE

The paper does not say QE has no benefits. It says that the liquidity benefit of QE is endogenous. A reserve injection can stimulate activity and reduce liquidity premia only if it increases *net usable liquidity.* If each reserve dollar induces a demandable deposit dollar, and if only part of the reserve dollar remains usable in stress, the marginal liquidity benefit is smaller than the balance-sheet expansion suggests. In bad parameter regions, the marginal effect can even go the other way.

7.4 Financial-stability implication

The model offers a theory of liquidity dependence. Repeated balance-sheet expansions may teach the private sector that central-bank liquidity will be available. Banks and nonbanks may then write more claims on liquidity: demandable deposits, credit lines, liquidity guarantees, and other runnable promises. The next liquidity stress then requires a larger intervention. This creates a ratchet: interventions stabilize the system today but can increase dependence tomorrow.

7.5 Monetary-policy implication

The most dangerous period may be after QT. QT reduces reserves, but it may not reduce the deposit base and credit commitments created during QE. If a shock hits during or after QT, the central bank may have to expand its balance sheet again even while trying to tighten monetary policy. This is the financial-dominance concern: liquidity support can interfere with anti-inflation policy because the banking system has become dependent on central-bank reserves.

7.6 Regulatory implication

The paper suggests that reserve adequacy should not be measured by reserves alone. Regulators should monitor the full liquidity ecosystem:

- uninsured demandable deposits;
- outstanding credit lines and liquidity commitments;
- reserve encumbrance and pledgeability;
- distribution of reserves across banks;
- incentives for safe banks to participate in interbank lending;
- expectations of ex post central-bank support.

A system with high reserves but high runnable claims may be less stable than a system with lower reserves but fewer claims and better interbank-market functioning.

7.7 What not to infer

The paper should not be read as a blanket argument against central-bank liquidity provision. In crises, central-bank intervention may be essential. The sharper point is that emergency liquidity and durable balance-sheet expansion have different incentive effects. Temporary liquidity can solve a run; repeated permanent reserve expansion can reshape private balance sheets in a way that makes future liquidity stress more likely or more severe.

7.8 Research agenda

The paper leaves several empirical and theoretical questions open. Future work could estimate how much QE-induced reserves are financed by uninsured demandable deposits, how quickly those deposits shrink during QT, how reserve distributions across banks affect interbank lending, and how expectations of central-bank intervention influence banks' credit-line supply. Another important direction is to connect this liquidity mechanism with interest-rate risk and bank solvency risk, especially after the 2023 regional-bank failures.

7.9 Takeaway

The deepest takeaway is that liquidity is a system property, not an asset label. A reserve is liquid for the bank holding it, but it is useful to the system only if it remains unencumbered and moves to where liquidity is needed. Central-bank balance sheet policy should therefore be evaluated by the net liquidity it creates after private-sector balance-sheet responses, not by the gross amount of reserves printed.